

# HR Studio Project Assessment

Current implementation and the path to a generalized MSME HR product

Engineering review | 15 September 2026 | MSME means micro, small and medium enterprise

## Executive assessment

**HR Studio is a useful interactive prototype, but it is not ready to run a company's HR operations.** Its strongest assets are the breadth of screens, shared employee data, consistent visual structure, editable letter templates and a small, understandable codebase. Its weakest areas are the foundations: records are temporary, users are not authenticated, payroll has no processing history, and several actions present simulated results as completed business operations. Retain the interface as a starting point; build reliable data and workflows before adding further modules.

## How the application is built

This is a TypeScript web application. React renders a single-page interface; Tailwind CSS provides styling, Motion supplies animation, Lucide supplies icons and Recharts draws the department chart. Vite serves and builds the frontend. A Node.js Express server serves the app and exposes four POST endpoints: `/api/email/send`, `/api/ai/draft`, `/api/ai/evaluate` and `/api/ai/chat`. Gemini handles optional text generation; Nodemailer handles optional SMTP email delivery. Firebase and Resend are dependencies, but neither is integrated into the application code.

**The main data path is form → module callback → App.tsx state → refreshed screen.** App.tsx owns employees, leave requests, attendance, document metadata, jobs, candidates, assets, appraisals, notifications and email logs. Child modules receive these arrays and callbacks through component properties. There are no HR record APIs, database connections, migrations or server-side repositories. Initial business-data arrays in data.ts are empty. A page reload resets the records; only the light or dark theme is saved in localStorage. Separate browsers do not share a company record.

## Identity and data relationships

Launching the portal changes a screen flag; it does not log anyone in. The displayed current user is employees[0], or a guest when the directory is empty. Adding an employee inserts them at the front, so it also changes who appears to be logged in and whose attendance is recorded. Job designation is a free-text employee field, not an access-control role. Everyone using the interface can reach the same administrative actions.

Employee is the central record: identity, contact, department, hire date, status and four salary amounts. Leaves, attendance, documents, assets and appraisals refer to employee IDs; candidates refer to job IDs. These are TypeScript shapes, not enforced database relationships. There is no organization, branch, reporting-manager, policy, leave-balance or payroll-run entity. Names are also copied into related records, so edits can leave old names behind; deleting an employee does not reconcile their related records.

## Scope and reading map

The review covers all 18 application source files, approximately 6,258 lines, plus the manifest, lockfile, configuration, environment example and README. App.tsx coordinates the app; types.ts defines the records; data.ts defines starting values; the components directory contains the screens; server.ts contains the external-service endpoints. Pages 2 and 3 explain the current behavior and its quality; page 4 prioritizes product improvements.

Code anchors: src/App.tsx:69-117, 311-339, 605-720; src/types.ts; src/data.ts; server.ts:182, 285, 336, 387; package.json. Source line numbers refer to the reviewed snapshot.

# What each module actually does

The following distinguishes implemented screen behavior from the business capability implied by its labels. All HR records below are held in the current browser session.

| Module               | Implemented behavior and present limits                                                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dashboard            | Counts staff, pending leave, active jobs and assigned assets; charts department distribution. Clock-in/out appends local logs for the displayed user. Celebrations are placeholder text; no birthday or anniversary schedule is calculated.                                                             |
| Employee directory   | Creates, edits, deletes, searches and filters profiles; calculates a salary breakdown. Departments are fixed choices. There are no worker-type, branch, manager, bank or statutory fields. Onboarding milestones are always marked complete, regardless of actual work.                                 |
| Leave and attendance | Accepts leave requests for selected staff, filters statuses and permits approval or rejection. Day counts use an absolute inclusive calendar difference, so reversed dates are accepted. No balances, accruals, holidays, overlap checks or shift rules. Approvals do not update attendance or payroll. |
| Payroll              | Calculates gross = basic + HRA + allowances, and net = gross - deductions; displays totals and a printable payslip. Changing the month changes its label, not the calculation history. No payroll run, approval, statutory calculation, bank transfer or payment reconciliation exists.                 |
| Documents            | Four editable local templates: offer, appointment, recommendation and experience letters. Copy and browser print are available. Upload and Save to Vault store a name/category/date and a random size; no file or draft content is stored, retrieved or downloaded. This screen does not call Gemini.   |
| ID badges            | Selects an employee, changes badge color, flips the design and invokes browser print. The apparent QR code is a fixed SVG pattern. There is no encoded employee link, RFID provisioning, verification endpoint or access-log integration.                                                               |
| Recruitment          | Creates job listings and renders a four-stage candidate board with a resume-scoring request. No applicant-entry form, resume upload, public application page or import exists; onAddCandidate is never invoked. Starting with empty data, the candidate workflow cannot be populated through the UI.    |
| Hardware assets      | Creates inventory records, searches equipment, changes custody/status and deletes assets. Assignment can accept an unknown employee ID. There is no custody history, return acknowledgment or warranty/cost record; purchase date is always the creation date.                                          |
| Email hub            | Builds birthday, anniversary, payslip and custom notices; supports copy, Gmail/mail-app drafts, previews and local logs. Its inbox is explicitly a simulator. The draft/copy actions also invoke server sending. No scheduler, real inbound mailbox, durable queue or receipt tracking exists.          |
| Performance          | Has review creation, self-rating, feedback, approval and averages in code. Its manager rating remains the default 4 because no form control changes it. Period choices are fixed. It has no reachable navigation entry in the current app.                                                              |
| Organization chart   | Groups staff by department and selects a VP/director by a title substring. It does not represent stored manager relationships. The component is wired into App.tsx but has no reachable navigation entry.                                                                                               |

**The hire-to-retire chain is incomplete.** Candidates do not become employees automatically; approved leave does not affect payable days; generated letters do not become stored files; payroll does not create a payment record or automatically dispatch a period-specific payslip. The landing-page workflow is an animated explanation, not a running workflow engine.

Evidence: src/components/modules/\*View.tsx; src/components/IDCardView.tsx; src/App.tsx:130-226, 343-354, 616-637; src/components/LandingPage.tsx:29-34.

# Engineering quality and key findings

## What is worth keeping

Feature-based components, typed records, clear callbacks and a lockfile make the project approachable. Forms, filters, local updates and empty states provide a useful base for product discovery. Secrets are read on the server, and Gemini requests use structured response schemas for drafting and scoring. The application can be evolved incrementally. Its interface is more mature than its data handling and business rules.

## Critical foundations and high priority defects

**1. No durable or protected HR system.** Reloading destroys data, and neither the UI nor the four endpoints authenticate users. If the server is reachable and configured, callers can submit arbitrary recipients to its mail service and invoke AI without per-user authorization or rate limits. Add identity, access rules, persistent storage and abuse controls before real employee use. HTML email templates also interpolate raw input, allowing untrusted markup to alter sent content. [App.tsx:92-117; server.ts:12, 128-178, 182-440]

**2. Communications have unintended effects and misleading statuses.** Copy Styled Email, Open in Gmail and Local Mail App all call the sending callback, potentially sending before the user submits a draft and duplicating a later manual send. Logs start as Sent before the request finishes. Ethereum is also marked Sent, although that service captures test mail without delivering to recipients. Its fallback can therefore send HR content to a testing service. Separate draft, queued, test, accepted, delivered and failed states. [EmailHubView.tsx:143-205, 339-373; App.tsx:234-275; [Nodemailer documentation](#)]

**3. Salary and document records are not trustworthy yet.** Payroll accepts generic deductions and permits negative amounts; all months use current employee salary. INR salary amounts become dollar amounts in letter templates without conversion. Email payslips are fixed to June 2026; zero salary components are replaced with sample amounts using `||` defaults. Cleared, transferred and compliant labels have no matching business evidence. Vault saving discards the actual draft. [PayrollView.tsx:24-39, 125-151; DocsView.tsx:33-95, 225-255; EmailHubView.tsx:477-495]

**4. Workflow integrity needs explicit rules.** IDs are random values from only 900 possibilities per record type, with no uniqueness check. Reversed leave dates become positive durations; attendance uses local time with a UTC date and does not restrict open sessions to today. Employee deletion can leave orphan records. Sidebar search changes tabs but never passes its query into the directory. These are correctness defects, not scale-only concerns. [App.tsx:154, 313, 321, 425-433, 616-648; LeavesView.tsx:59-62]

**5. AI capability is narrower than advertised.** Draft and chat endpoints have no UI callers. Resume scoring is reachable only when candidate data exists; the UI ignores HTTP failure status and simulation flags, and a network exception fabricates a score of 75. Missing-key server scoring returns 65 or 88 based on two keywords. Chat uses hardcoded, internally inconsistent handbook text; it does not read company policy documents. Real model availability and output quality remain unverified. [RecruitmentView.tsx:69-96; server.ts:260-439]

## Maintainability and validation

App.tsx is 730 lines and EmailHubView.tsx is 998 lines. Repeated templates and salary formulas already diverge. There is no test suite, CI pipeline, lint-rule configuration or strict TypeScript setting; the lint script only type-checks. The README is starter documentation: it recommends `.env.local`, while `server dotenv.config()` loads `.env` by default. Start requires `NODE_ENV=production` to use built assets; the script does not set it. Port 3000 is fixed, and production static serving includes the generated server bundle and source map in `dist`. [package.json; tsconfig.json; README.md; server.ts:7-10, 442-460]

Validation: TypeScript check and production build passed; the initial JavaScript bundle is 888 kB (251 kB gzip), with a size warning. Local checks confirmed portal access without login, guest attendance lost on reload, unauthenticated simulated AI scoring, and public `server-bundle/source-map` URLs. No live email or paid AI calls were made. This local-snapshot review is not a deployment, load-test or full security audit.

# Improvements for a generalized MSME product

**Product direction:** build one configurable HR core with optional modules and country-specific payroll packs. India is a proposed first market, inferred from the INR/HRA interface and the MSME brief; the target geography is not confirmed. Office, retail, workshop and field teams need different policies while sharing the same employee lifecycle.

## Priority 0 Make existing operations reliable

Add organizations, users, memberships and explicit owner, HR, manager, payroll and employee permissions. Store business records in a relational database with enforced relationships, unique IDs, migrations, organization-scoped queries and restore-tested backups. Use private file storage with access-controlled downloads. A modular Express service plus PostgreSQL is a reasonable evolution of this code; row-level policies can add database-level separation alongside application checks. [[PostgreSQL row security](#)]

Resolve the defects on page 3, separate email drafting from sending, remove unsupported completion/security claims and distinguish demo mode visibly. Add server-side validation, escaped templates, request limits, audit history and alerts for external-service spending. Split frontend assets from server build output. Add automated checks for permissions, persistence, record relationships, leave dates and salary arithmetic. [[OWASP API resource controls](#)]

**Exit condition:** two users can work with the same saved company records; another company cannot access them; every salary or approval change records who changed it; document retrieval and backup restoration succeed.

## Priority 1 Complete the daily MSME workflow

**Company setup and employee self-service.** Configure legal entities, branches, departments, managers, cost centers, holidays, timezone, currency, branding and custom fields. Provide validated spreadsheet import and export, employee invitations, profile-change approvals, emergency contacts, bank details and worker categories such as salaried, hourly, daily-wage and contract. Keep the mobile experience simple and usable on limited bandwidth.

**Attendance, leave and payroll.** Add shifts including overnight work, weekly offs, approved overtime, attendance corrections, device/CSV imports, leave accruals and balances, and approval delegation. Feed approved payable days into payroll. Store effective-dated salary structures and immutable monthly runs with proration, loss of pay, advances, reimbursements, arrears, rounding, review/lock steps, bank exports and reconciliation. Earlier payslips must remain unchanged after a salary revision.

For an India pack, plan configurable EPF/ESI, income-tax withholding, professional tax and labour-welfare-fund support where applicable, plus statutory exports and a compliance calendar. Validate eligibility, rates and effective dates against the relevant jurisdiction before release. The EPFO employer portal illustrates the need for structured, UAN-based ECR output; a single deductions number cannot represent this workflow. [[EPFO employer portal](#)]. These are proposed product capabilities, not a compliance certification.

**Exit condition:** a pilot company completes an attendance-to-payroll cycle, reconciles totals against its approved reference payroll, and lets each employee securely retrieve the correct payslip.

## Priority 2 Extend the employee lifecycle

Complete applicant intake, interviews, offer approvals and candidate-to-employee conversion. Add task-based onboarding, probation reminders, asset issue/return acknowledgments, resignations, clearance and final settlement. Introduce expenses and claims, policy acknowledgments, HR requests, useful reports and accounting integrations. Add reliable scheduled notices with retries and duplicate prevention. Reconnect performance and the organization chart after manager relationships and review permissions exist.

Use AI for reviewed drafts and cited answers from approved company policies, with permission-aware access and data minimization. Candidate scores should have transparent criteria and human review; failed calls must remain failures. Offer sector presets through configuration, then validate them with representative MSMEs. Prioritize successful real workflows over additional dashboard screens.